

Lecture 08 - Introduction

Lecture 08 - Hypothesis Testing - Introduction

Purpose of the Lecture

This lecture introduces hypothesis testing. The purpose is to use sample data to evaluate a claim about an unknown population or process.

The previous lectures developed the ingredients needed for this step. We studied distributions, sampling fluctuation, standard errors, and confidence intervals. Hypothesis testing uses the same ideas, but organizes them around a decision. Instead of asking only which values are plausible, we start with a benchmark claim and ask whether the observed sample evidence is strong enough to challenge it.

The lecture begins with the logic of a test. We define a null hypothesis, which represents the benchmark claim, and an alternative hypothesis, which represents the departure that matters for the problem. The sample is then evaluated under the assumption that the benchmark is true. If the sample result would be unusually far from what the benchmark predicts, we reject the null hypothesis. Otherwise, we fail to reject it.

The lecture also connects hypothesis tests to confidence intervals. For simple two-sided tests, the logic is closely related: when a confidence interval excludes a benchmark value, the corresponding test rejects that benchmark at the associated significance level. This link helps students see hypothesis testing as an extension of uncertainty measurement rather than a separate mechanical procedure.

The second part of the lecture applies the full procedure to a call-center example. A company claims that average customer waiting time is at most 90 seconds. A sample of calls gives an average waiting time of 93 seconds. The question is whether this difference is large enough, relative to sampling variation, to reject the company's claim.

Learning Objectives

This lecture aims to:

1. Explain the purpose of hypothesis testing.
2. Define the null hypothesis and the alternative hypothesis.
3. Distinguish one-sided and two-sided tests.
4. Explain the significance level and its connection to Type I error.
5. Distinguish Type I and Type II errors.
6. Compute a standard error for a sample mean.
7. Compute a test statistic as evidence relative to sampling variability.
8. Use critical values for right-tail, left-tail, and two-sided tests.
9. Interpret p-values correctly.
10. Write statistical conclusions in careful business language.

Learning Outcomes

By the end of this session, students should be able to:

1. Translate a business or finance claim into null and alternative hypotheses.
2. Decide whether a test should be one-sided or two-sided.
3. Explain why the direction of the alternative must come from the problem, not from the observed result.
4. Compute a z-statistic for a sample mean in a simple large-sample setting.
5. Compare a test statistic with a critical value.
6. Compare a p-value with a significance level.
7. State whether to reject or fail to reject the null hypothesis.
8. Explain why failing to reject the null does not prove that the null is true.
9. Distinguish statistical significance from practical importance.
10. Write a conclusion that connects the statistical decision to the original context.

Lecture Roadmap

The lecture starts by moving from confidence intervals to tests. Confidence intervals report a range of plausible parameter values. Hypothesis tests focus on one benchmark value or one benchmark claim.

Next, the lecture defines the two hypotheses. The null hypothesis is the benchmark. The alternative hypothesis is the competing claim. The lecture uses simple examples, such as a call center claiming that average waiting time is at most 90 seconds, to show how the hypotheses are formed.

The lecture then introduces the main vocabulary: one-sided tests, two-sided tests, Type I error, Type II error, and significance level. These concepts are then assembled into a five-step procedure.

After that, the lecture explains the role of the test statistic. A raw difference between a sample estimate and a benchmark is not enough. We need to compare that difference with the standard error. The test statistic measures how far the sample estimate is from the benchmark in standard-error units.

The lecture then studies critical values for right-tail, left-tail, and two-sided tests. It also introduces p-values as an alternative way to summarize evidence against the null hypothesis.

The final part gives a full worked example. A call center claims average waiting time is at most 90 seconds. A random sample of 100 calls gives a mean of 93 seconds and a standard deviation of 10 seconds. The example shows how to formulate the hypotheses, compute the standard error, calculate the z-statistic, compare it with the critical value, and write the conclusion.

Central Idea

The central idea of the lecture is:

A hypothesis test asks whether the sample evidence is unusually far from what a benchmark claim would predict.

For example, a sample mean of 93 seconds is above a benchmark of 90 seconds. By itself, this difference does not prove anything. Samples fluctuate. The difference becomes meaningful only after it is compared with the standard error. If the standard error is 1 second, then 93 is three standard errors above 90. That is much stronger evidence than a difference of 3 seconds with a standard error of 5 seconds.

How to Work Through the Lecture

Begin by identifying the claim being tested. Translate that claim into a null hypothesis. Then identify the alternative hypothesis from the practical question.

Next, determine the direction of the test. A one-sided test is appropriate when only one direction is relevant. A two-sided test is appropriate when deviations in both directions matter.

Then compute the standard error and the test statistic. The key habit is to compare the observed estimate with the benchmark relative to sampling variability.

Finally, write the conclusion in context. Avoid saying that the test proves a hypothesis. Use language such as "the data provide strong evidence against the benchmark" or "the sample does not provide strong enough evidence to reject the benchmark."

What to Pay Attention To

Three habits are especially important.

First, choose the alternative hypothesis before looking at the result. The direction of the test should come from the business or scientific question.

Second, interpret p-values correctly. A p-value is not the probability that the null hypothesis is true. It is a probability about the sample evidence, computed under the null model.

Third, separate statistical significance from practical importance. A statistically significant result may be too small to matter in a business setting. A non-significant result may still be practically interesting if the data are noisy or the sample is small.

End-of-Lecture Check

By the end of the lecture, you should be comfortable answering questions such as:

1. What is the null hypothesis?
2. What is the alternative hypothesis?
3. What makes a test one-sided or two-sided?
4. What does the significance level control?
5. What is a Type I error?
6. What is a Type II error?
7. Why is the raw difference between an estimate and a benchmark not enough?
8. What does a test statistic measure?
9. What is a p-value?
10. How should we write a careful conclusion after a hypothesis test?
11. If these questions are clear, you are ready to use hypothesis testing as a foundation for regression, beta estimation, and broader empirical decision-making.

If these questions are clear, you are ready to use confidence intervals as the foundation for hypothesis testing and regression inference.